11. Income Taxes

Tax Cuts and Jobs Act

In December 2017, the Tax Cuts and Jobs Act, or the Tax Act ("TCJA"), was signed into law. Among other things, the Tax Act permanently lowered the corporate federal income tax rate to 21% from the existing maximum rate of 35%, effective for tax years including or commencing January 1, 2018. As a result of the reduction of the corporate federal income tax rate GAAP required companies to revalue their deferred tax assets and deferred tax liabilities as of the date of enactment, with the resulting tax effects accounted for in the reporting period of enactment. For the year ending December 31, 2017, this revaluation resulted in a provision of $10,609 to income tax expense in continuing operations and a corresponding reduction the Company's valuation allowance. There was no impact, therefore, to the Company's income statement for the year ended December 31, 2017 as a result of the reduction in federal income tax rates. During 2018 the Company completed its review of new tax regulations issued under TCJA and finalized adjustments related to its deferred tax assets and liabilities that resulted from changes in the tax law.

Income Taxes and Deferred Tax Assets and Liabilities

The components of loss from operations before provision for income taxes consist of the following:

	Year Ended December 31,	
	2018	2017
Domestic	$ (12,273)	$ (9,523)
Foreign	3,103	126
Loss before taxes	$ (9,170)	$ (9,397)

Table of Contents

Significant components of the Company's net deferred tax assets are as follows:

	Year Ended December 31,	
	2018	2017
Deferred Tax Assets:		
Net operating loss carryforward	$ 24,261	$ 20,490
Capitalization of research and development expense	1,385	1,606
Credit carryforwards	2,664	2,493
Depreciation	—	990
Stock compensation	966	1,035
Other temporary differences	695	794
Total deferred tax assets.	29,971	27,408
Valuation allowance	(29,672)	(27,408)
Net deferred tax assets	299	—
Deferred Tax Liabilities:		
Depreciation	(299)	—
Net deferred taxes	$ —	$ —

Tax Rate

The items accounting for the difference between the income tax computed at the federal statutory rate of 21% and the provision for income taxes were as follows:

	Year Ended December 31,	
	2018	**2017**
Federal income tax at statutory federal rate	21.0 %	34.0 %
State taxes	8.0 %	4.9 %
Permanent differences	(0.1)%	(2.3)%
Tax credits	3.1 %	3.7 %
Federal rate change under tax reform	0.0 %	(112.9)%
State rate change on deferred balances	0.0 %	0.1 %
Stock compensation	(3.6)%	(12.7)%
Other	0.6 %	1.2 %
Change in valuation allowance	(25.2)%	84.0 %
German subsidiary dissolution	(3.8)%	0.0 %
Total	0.0 %	0.0 %

Tax Attributes

At December 31, 2018, the Company had U.S. net operating loss carryforwards (NOLs) for federal and state income tax purposes of approximately $89,971 and $84,918, respectively. The Company's existing federal and state operating loss carryforwards will begin to expire on various dates through 2038. Included in the $89,971 of federal net operating losses are losses of $12,162 that will carry forward indefinitely as a result of new regulations issued under TCJA. The Company also had available research and development and investment tax credits for federal and state income tax purposes of approximately $1,382 and $846, respectively. These federal and state research and development credits will begin to expire on various dates through 2038. In Canada, the Company has cumulative research tax credits totaling $614 that may be carried forward up to twenty years from the year earned. These credits will begin to expire in 2032 if not used. Management

of the Company has evaluated the positive and negative evidence bearing upon the realizability of its deferred tax assets, which are comprised principally of net operating loss carryforwards and research and development credits. Under the applicable accounting standards, management has considered the Company's history of losses and concluded that it is more likely than not that the Company will not recognize the benefits of federal and state deferred tax assets. Accordingly, a full valuation allowance has been established against the deferred tax assets.

Utilization of the net operating loss and research and development credit carryforwards may be subject to a substantial annual limitation under Section 382 of the Internal Revenue Code of 1986 due to ownership change limitations that have occurred previously or that could occur in the future. These ownership changes may limit the amount of net operating loss and research and development credit carryforwards that can be utilized annually to offset future taxable income and tax, respectively. The Company completed an evaluation of its ownership changes through December 31, 2015 and determined that an ownership change occurred on August 22, 2014 in connection with an equity offering. As a consequence of this ownership change, the Company's NOLs, tax credit carryforwards and other tax deductions allocable to the tax periods preceding the ownership change became subject to limitation under Section 382. The Company has reduced its associated deferred tax assets accordingly. The Company has not yet

completed an evaluation of ownership changes for the years 2016 through 2018. To the extent an ownership change occurs in the future, the net operating loss, credit carryforwards and other deferred tax assets may be subject to further limitations.

Other

During the year ended December 31, 2018, the Company decided to dissolve its wholly-owned German subsidiary, Metabolix GmbH, that has been inactive since 2014. As a result of this decision, the Company has written off the German deferred tax assets and related full valuation allowance resulting in no impact to the tax provision. The majority of the deferred tax asset value related to net operating loss carryforwards.

The tax years 2014 through 2017 remain open to examination by major taxing jurisdictions to which the Company is subject, which are primarily in the U.S. The statute of limitations for net operating losses utilized in future years will remain open beginning in the year of utilization.

The Company's policy is to record estimated interest and penalties related to uncertain tax positions as income tax expense. As of December 31, 2018 and 2017, the Company had no accrued interest or penalties recorded related to uncertain tax positions.

No additional provision has been made for U.S. income taxes related to the undistributed earnings of the wholly-owned subsidiaries of Yield10 Bioscience, Inc. or for unrecognized deferred tax liabilities for temporary differences related to investments in subsidiaries as the amounts are not significant. As such, earnings are expected to be permanently reinvested, the investments are essentially permanent in duration, or the Company has concluded that no additional tax liability will arise as a result of the distribution of such earnings. A liability could arise if amounts are distributed by such subsidiaries or if such subsidiaries are ultimately disposed. It is not practical to estimate the additional income taxes related to permanently reinvested earnings or the basis differences related to investment in subsidiaries. Unremitted earnings at December 31, 2018 and December 31, 2017 approximated $491 and $482, respectively.